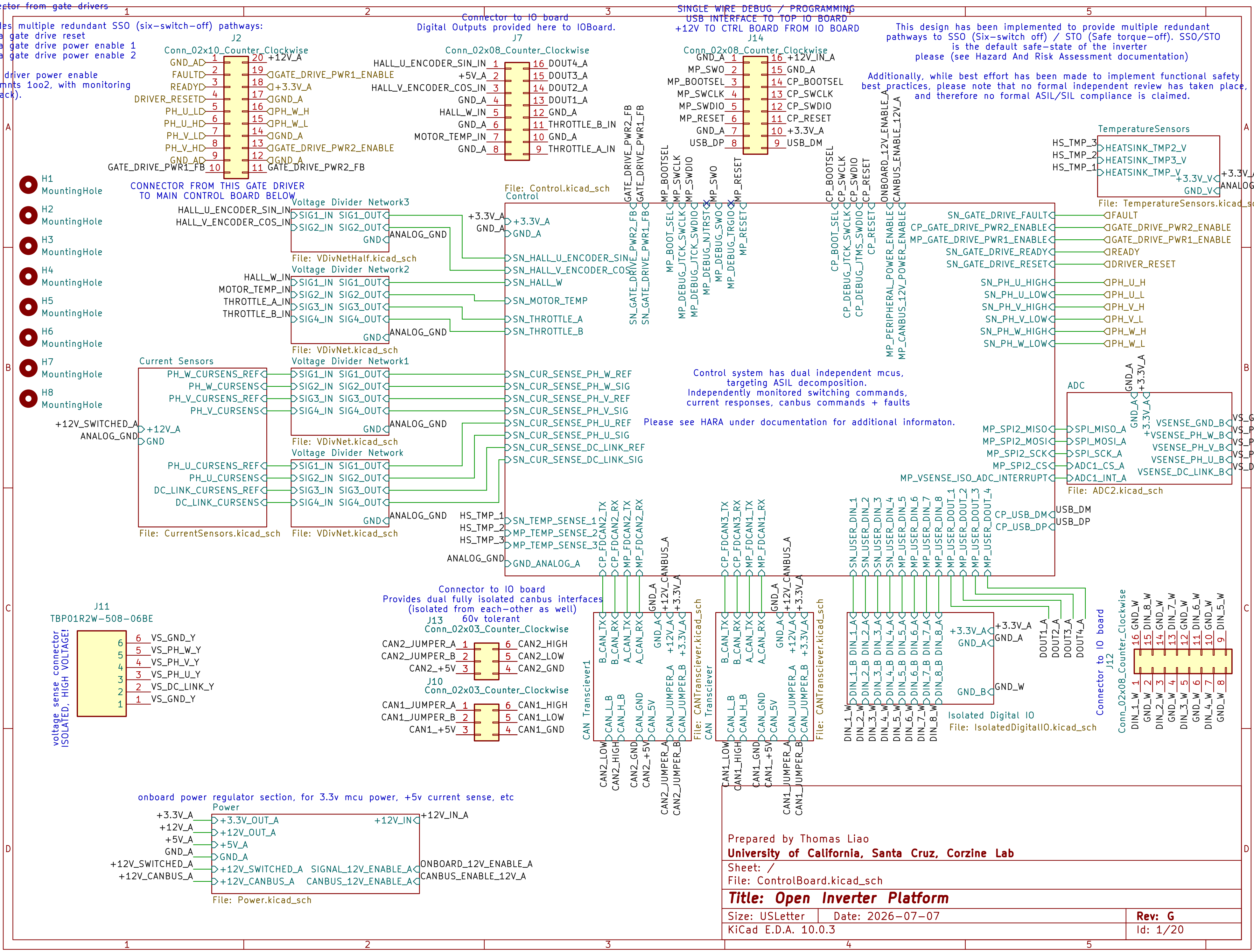
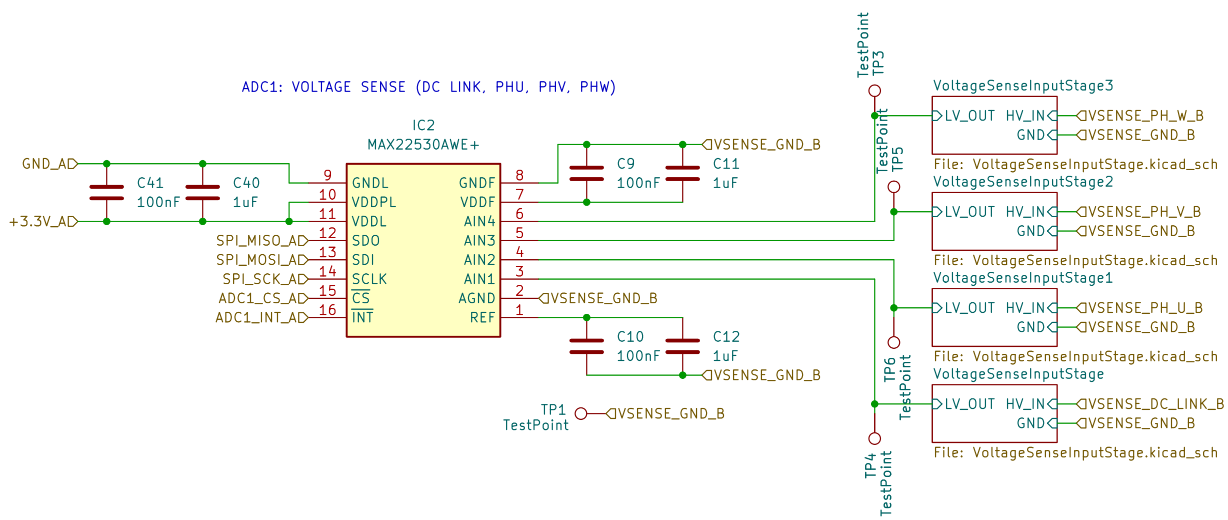

Connector from gate drivers
Includes multiple redundant SSO (six-switch-off) pathways:
– via gate drive reset
– via gate drive power enable 1
– via gate drive power enable 2
(Gate driver power enable implemnts 1oo2, with monitoring feedback).
J2
Conn_02x10_Counter_Clockwise
CONNECTOR FROM THIS GATE DRIVER TO MAIN CONTROL BOARD BELOW
Connector to IO board
Digital Outputs provided here to IOBoard.
J7
Conn_02x08_Counter_Clockwise
SINGLE WIRE DEBUG / PROGRAMMING
USB INTERFACE TO TOP IO BOARD
+12V TO CTRL BOARD FROM IO BOARD
J14
Conn_02x08_Counter_Clockwise
This design has been implemented to provide multiple redundant pathways to SSO (Six-switch off) / STO (Safe torque-off). SSO/STO is the default safe-state of the inverter please (see Hazard And Risk Assessment documentation)
Additionally, while best effort has been made to implement functional safety best practices, please note that no formal independent review has taken place, and therefore no formal ASIL/SIL compliance is claimed.
TemperatureSensors
File: TemperatureSensors.kicad_sch
H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole
H5 MountingHole
H6 MountingHole
H7 MountingHole
H8 MountingHole
Voltage Divider Network3
File: VDivNetHalf.kicad_sch
Voltage Divider Network2
File: VDivNet.kicad_sch
Voltage Divider Network1
File: VDivNet.kicad_sch
Voltage Divider Network
File: VDivNet.kicad_sch
Current Sensors
File: CurrentSensors.kicad_sch
Control
File: Control.kicad_sch
Control system has dual independent mcus, targeting ASIL decomposition. Independently monitored switching commands, current responses, canbus commands + faults
Please see HARA under documentation for additional informaton.
ADC
File: ADC2.kicad_sch
J11
TBP01R2W-508-06BE
voltage sense connector ISOLATED, HIGH VOLTAGE!
Connector to IO board
Provides dual fully isolated canbus interfaces (isolated from each-other as well) 60v tolerant
J13
Conn_02x03_Counter_Clockwise
J10
Conn_02x03_Counter_Clockwise
CAN Transciever1
File: CANTransciever.kicad_sch
CAN Transciever
File: CANTransciever.kicad_sch
Isolated Digital IO
File: IsolatedDigitalIO.kicad_sch
Connector to IO board
J12
Conn_02x08_Counter_Clockwise
onboard power regulator section, for 3.3v mcu power, +5v current sense, etc
Power
File: Power.kicad_sch
Prepared by Thomas Liao
University of California, Santa Cruz, Corzine Lab
Sheet: /
File: ControlBoard.kicad_sch
Title: Open Inverter Platform
Size: USLetter
Date: 2026-07-07
Rev: G
KiCad E.D.A. 10.0.3
Id: 1/20



Prepared by Thomas Liao

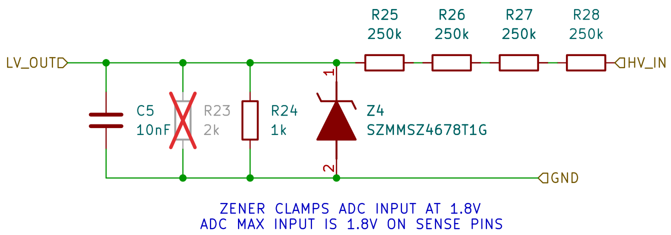
University of California, Santa Cruz, Corzine Lab

Sheet: /ADC/

File: ADC2.kicad_sch

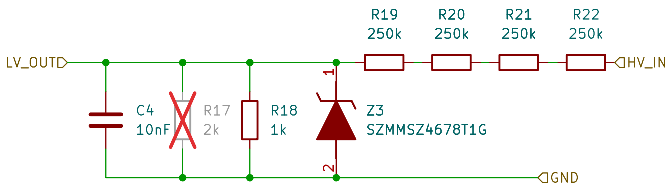
Title: Open Inverter Platform

Size: USLetter	Date: 2026-07-07	**Rev: G**
KiCad E.D.A. 10.0.3		Id: 4/20



Prepared by Thomas Liao
University of California, Santa Cruz, Corzine Lab
Sheet: /ADC/VoltageSenseInputStage/
File: VoltageSenseInputStage.kicad_sch
Title: Open Inverter Platform

Size: USLetter	Date: 2026-07-07	**Rev: G**
KiCad E.D.A. 10.0.3		Id: 2/20



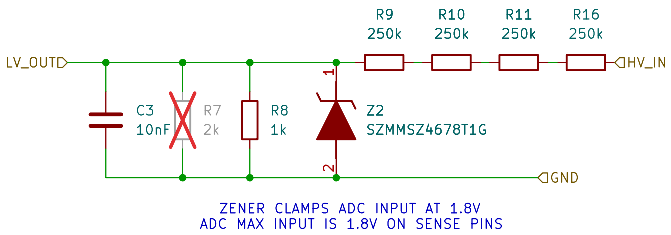
ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

Prepared by Thomas Liao
University of California, Santa Cruz, Corzine Lab

Sheet: /ADC/VoltageSenseInputStage1/
File: VoltageSenseInputStage.kicad_sch

Title: Open Inverter Platform

Size: USLetter	Date: 2026-07-07	**Rev: G**
KiCad E.D.A. 10.0.3		Id: 3/20

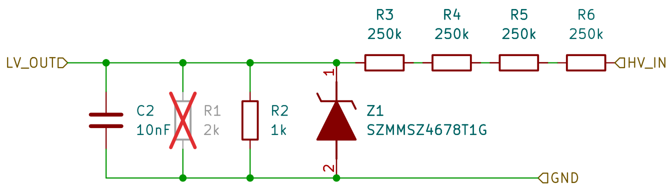


Prepared by Thomas Liao
University of California, Santa Cruz, Corzine Lab

Sheet: /ADC/VoltageSenseInputStage2/
File: VoltageSenseInputStage.kicad_sch

Title: Open Inverter Platform

Size: USLetter | Date: 2026-07-07 | **Rev: G**
KiCad E.D.A. 10.0.3 | Id: 5/20



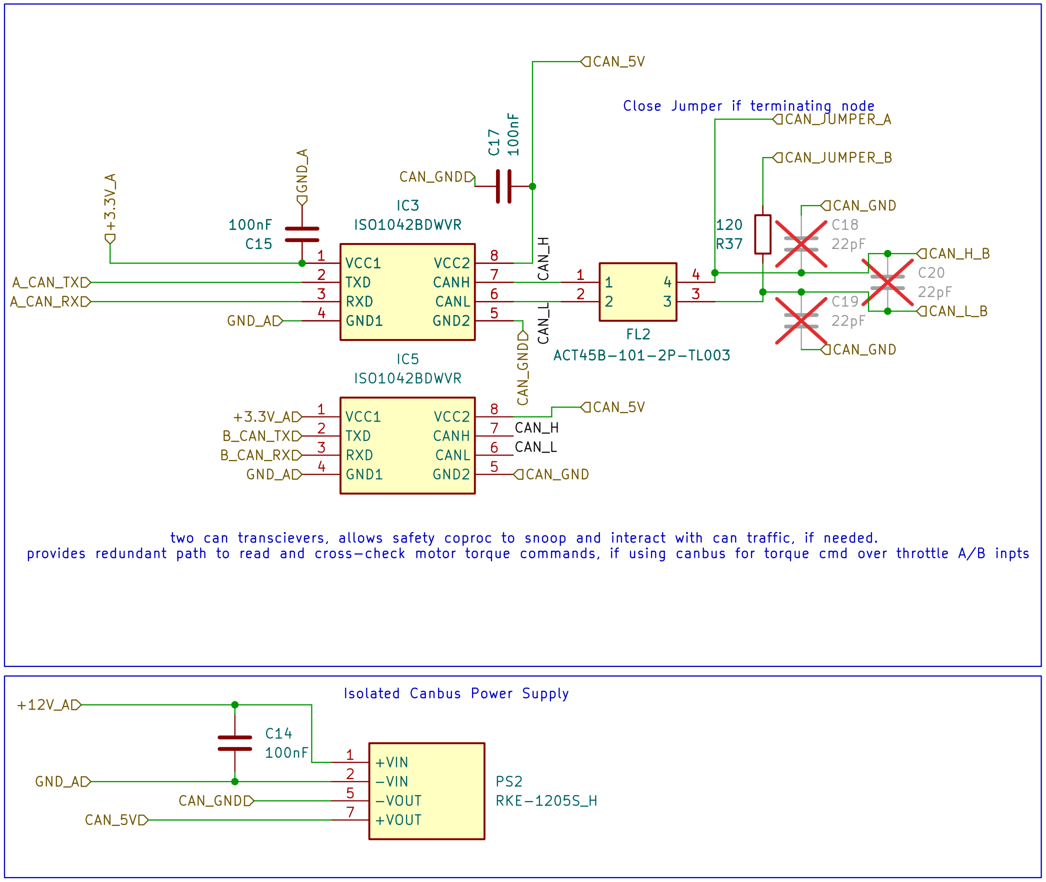
ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

Prepared by Thomas Liao
University of California, Santa Cruz, Corzine Lab

Sheet: /ADC/VoltageSenseInputStage3/
File: VoltageSenseInputStage.kicad_sch

Title: Open Inverter Platform

Size: USLetter	Date: 2026-07-07	**Rev: G**
KiCad E.D.A. 10.0.3		Id: 6/20



Prepared by Thomas Liao

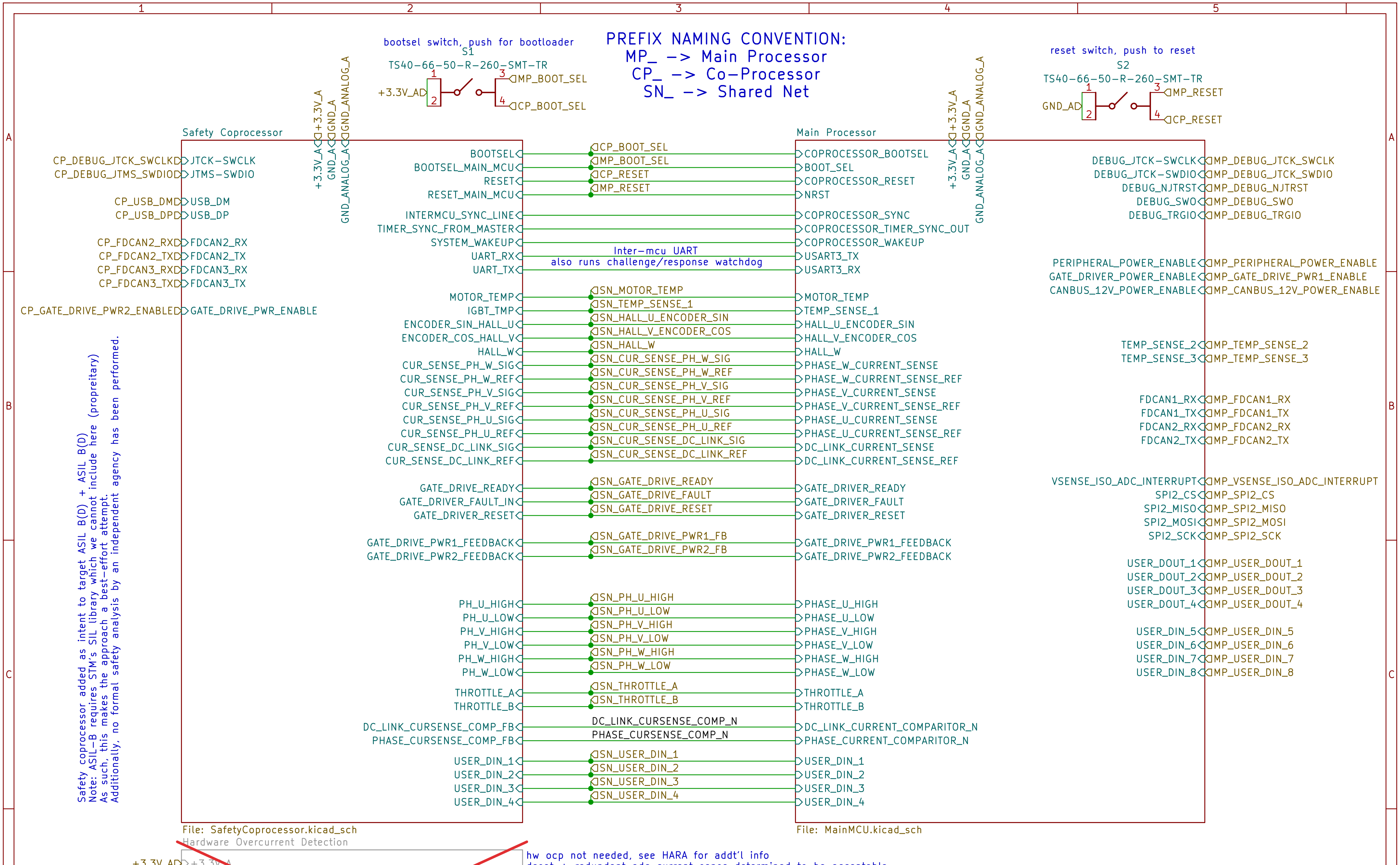
University of California, Santa Cruz, Corzine Lab

Sheet: /CAN Transciever1/

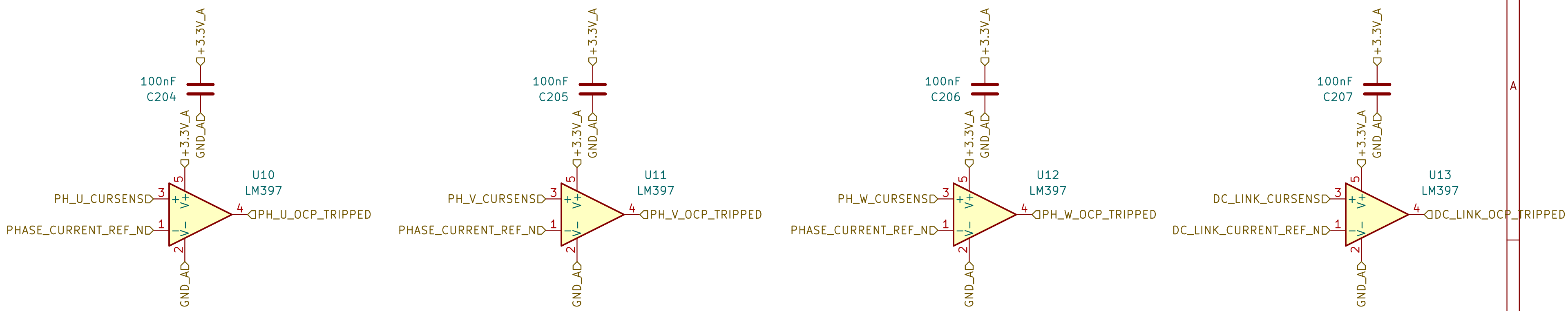
File: CANTransciever.kicad_sch

Title: Open Inverter Platform

Size: USLetter	Date: 2026-07-07	**Rev: G**
KiCad E.D.A. 10.0.3		Id: 7/20



Prepared by Thomas Liao
University of California, Santa Cruz, Corzine Lab
Sheet: /Control/
File: Control.kicad_sch
Title: Open Inverter Platform
Size: USLetter | Date: 2026-07-07 | **Rev: G**
KiCad E.D.A. 10.0.3 | Id: 8/20

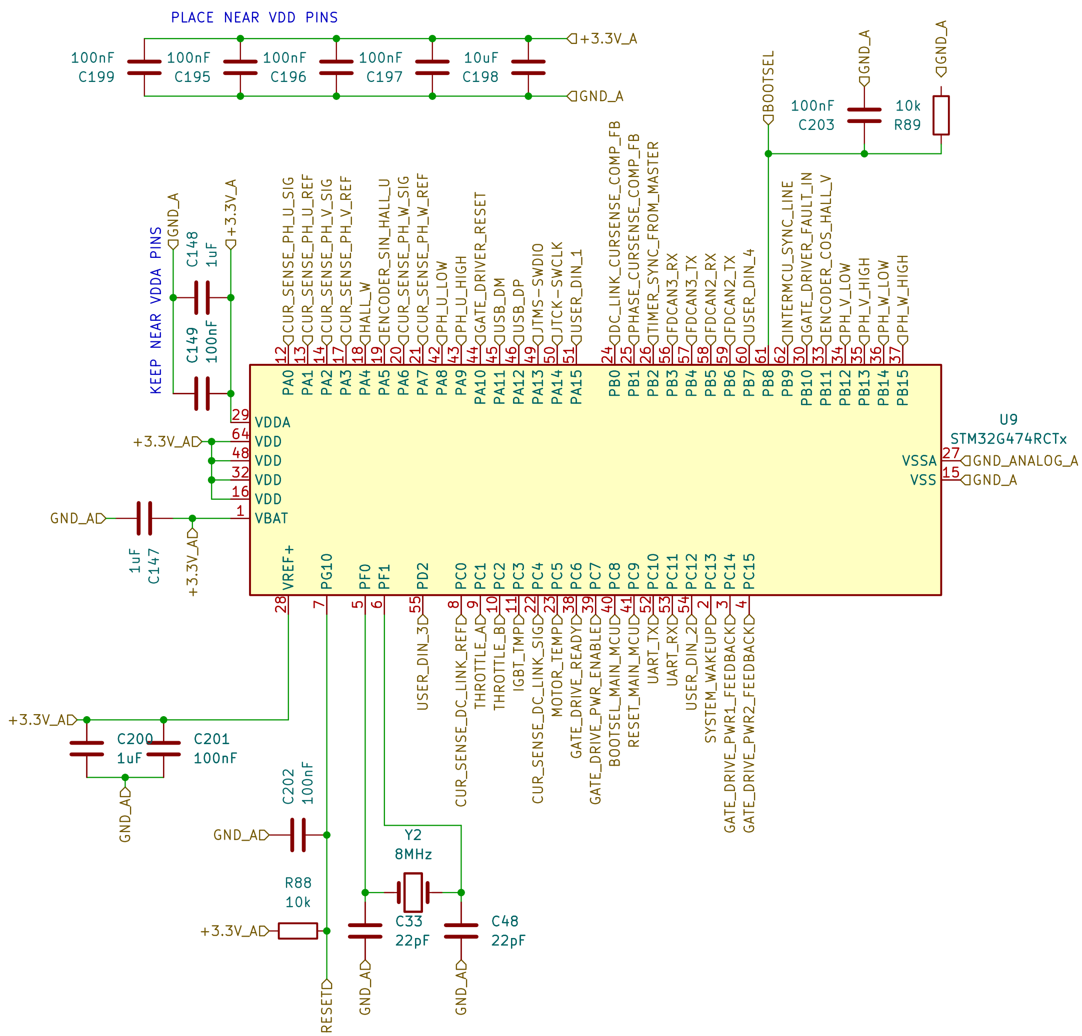


Prepared by Thomas Liao
University of California, Santa Cruz, Corzine Lab

Sheet: /Control/Hardware Overcurrent Detection/
File: HWOCP.kicad_sch

Title: Open Inverter Platform

Size: USLetter	Date: 2026-07-07	**Rev: G**
KiCad E.D.A. 10.0.3		Id: 10/20



Prepared by Thomas Liao

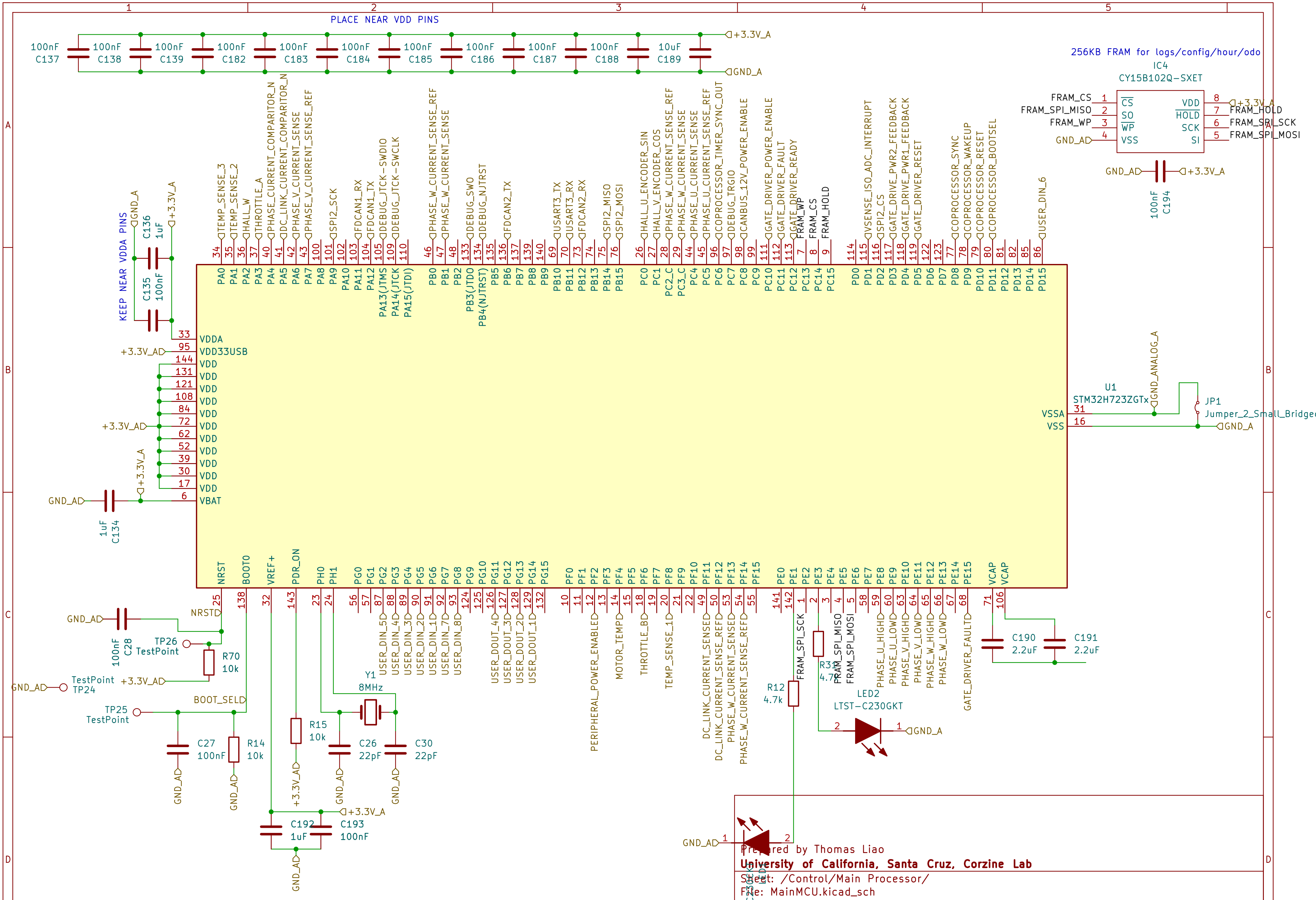
University of California, Santa Cruz, Corzine Lab

Sheet: /Control/Safety Coprocessor/

File: SafetyCoprocessor.kicad_sch

Title: Open Inverter Platform

Size: USLetter	Date: 2026-07-07	**Rev: G**
KiCad E.D.A. 10.0.3		Id: 16/20



Prepared by Thomas Liao

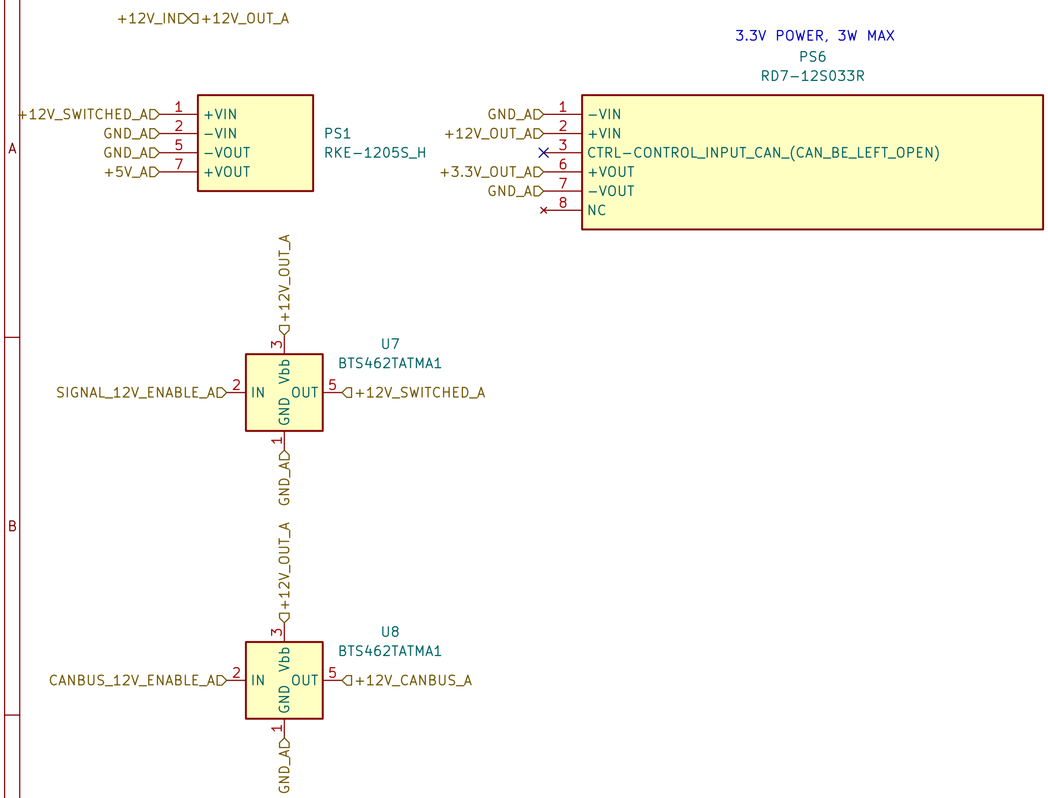
University of California, Santa Cruz, Corzine Lab

Sheet: /Control/Main Processor/

File: MainMCU.kicad_sch

Title: Open Inverter Platform

Size: USLetter	Date: 2026-07-07	Rev: G
KiCad E.D.A. 10.0.3		Id: 19/20



Prepared by Thomas Liao

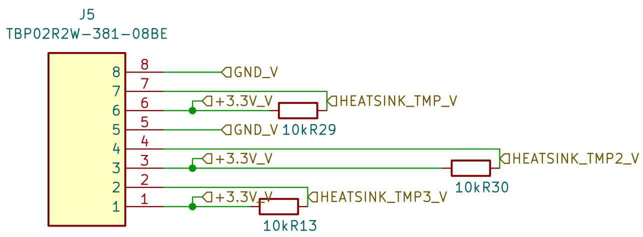
University of California, Santa Cruz, Corzine Lab

Sheet: /Power/

File: Power.kicad_sch

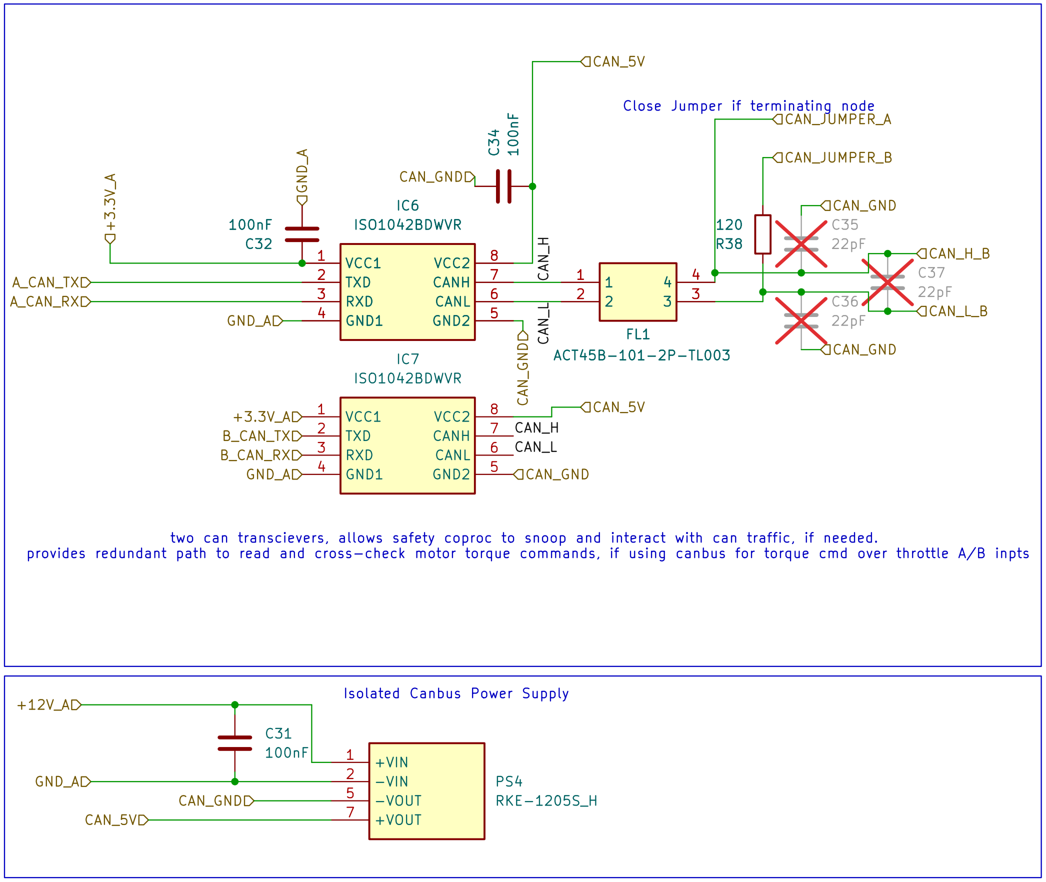
Title: Open Inverter Platform

Size: USLetter	Date: 2026-07-07	**Rev: G**
KiCad E.D.A. 10.0.3		Id: 9/20



Prepared by Thomas Liao
University of California, Santa Cruz, Corzine Lab
Sheet: /TemperatureSensors/
File: TemperatureSensors.kicad_sch
Title: Open Inverter Platform

Size: USLetter	Date: 2026-07-07	**Rev: G**
KiCad E.D.A. 10.0.3		Id: 9/20



Prepared by Thomas Liao

University of California, Santa Cruz, Corzine Lab

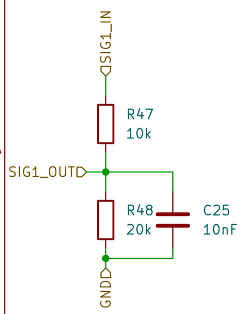
Sheet: /CAN Transciever/

File: CANTransciever.kicad_sch

Title: Open Inverter Platform

Size: USLetter | Date: 2026-07-07 | **Rev: G**

KiCad E.D.A. 10.0.3 | Id: 11/20



Prepared by Thomas Liao

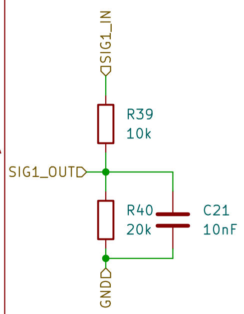
University of California, Santa Cruz, Corzine Lab

Sheet: /Voltage Divider Network1/

File: VDivNet.kicad_sch

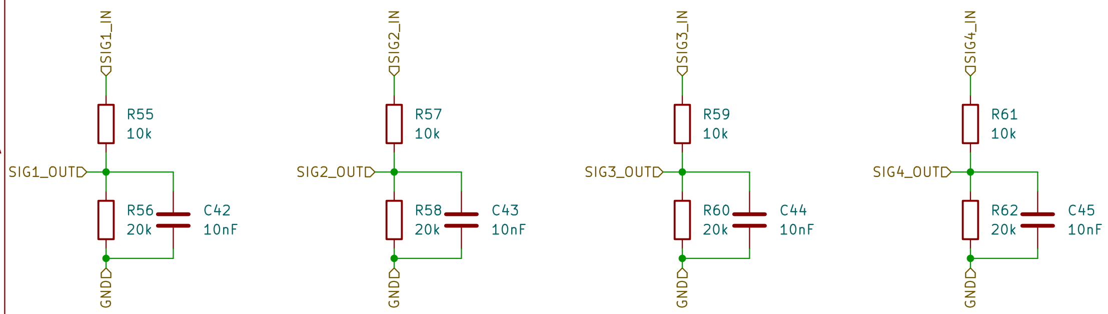
Title: Open Inverter Platform

Size: USLetter	Date: 2026-07-07	**Rev: G**
KiCad E.D.A. 10.0.3		Id: 12/20



Prepared by Thomas Liao
University of California, Santa Cruz, Corzine Lab
Sheet: /Voltage Divider Network/
File: VDivNet.kicad_sch
Title: Open Inverter Platform

Size: USLetter	Date: 2026-07-07	**Rev: G**
KiCad E.D.A. 10.0.3		Id: 13/20

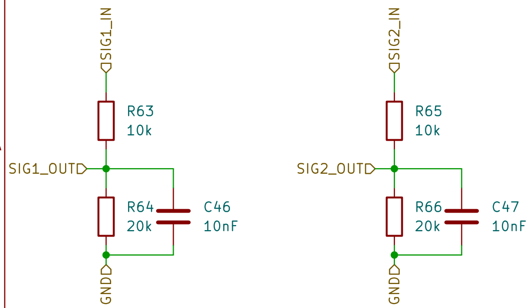


Prepared by Thomas Liao
University of California, Santa Cruz, Corzine Lab

Sheet: /Voltage Divider Network2/
File: VDivNet.kicad_sch

Title: Open Inverter Platform

Size: USLetter	Date: 2026-07-07	**Rev: G**
KiCad E.D.A. 10.0.3		Id: 14/20



Prepared by Thomas Liao

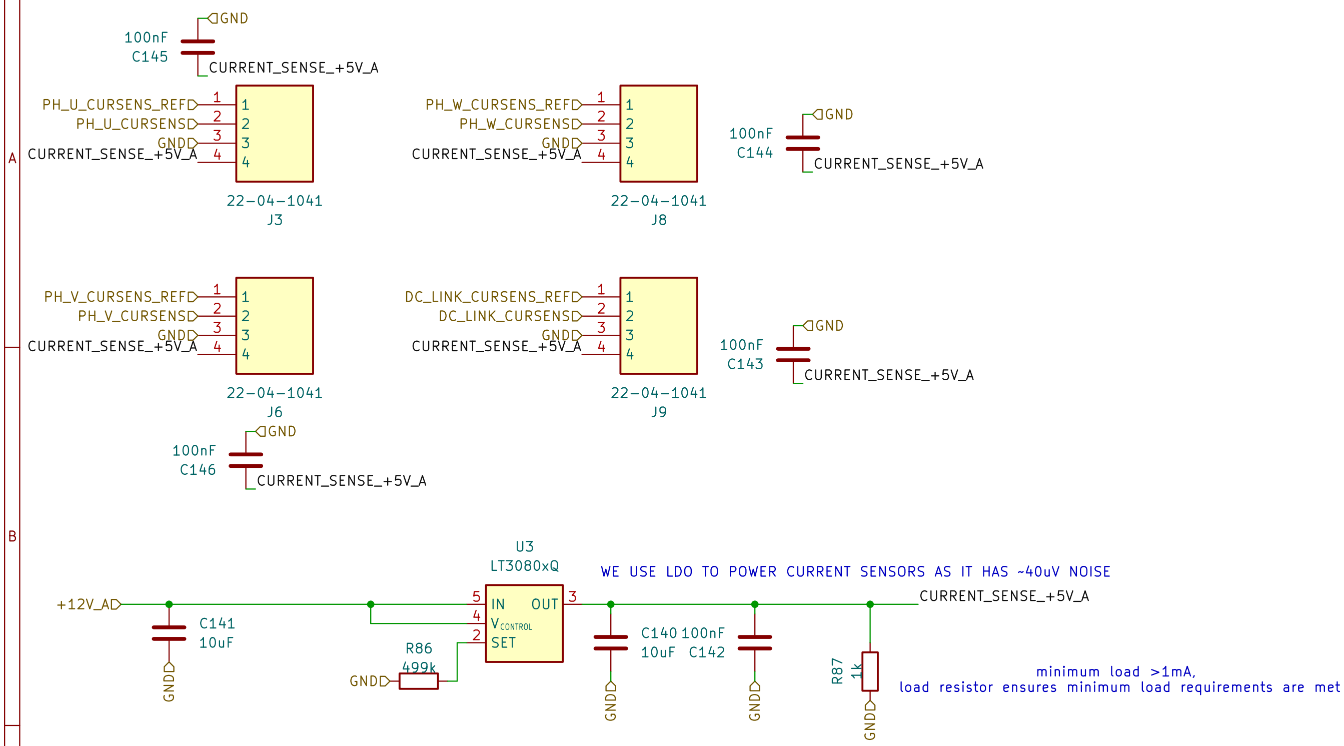
University of California, Santa Cruz, Corzine Lab

Sheet: /Voltage Divider Network3/

File: VDivNetHalf.kicad_sch

Title: Open Inverter Platform

Size: USLetter	Date: 2026-07-07	**Rev: G**
KiCad E.D.A. 10.0.3		Id: 15/20



Prepared by Thomas Liao

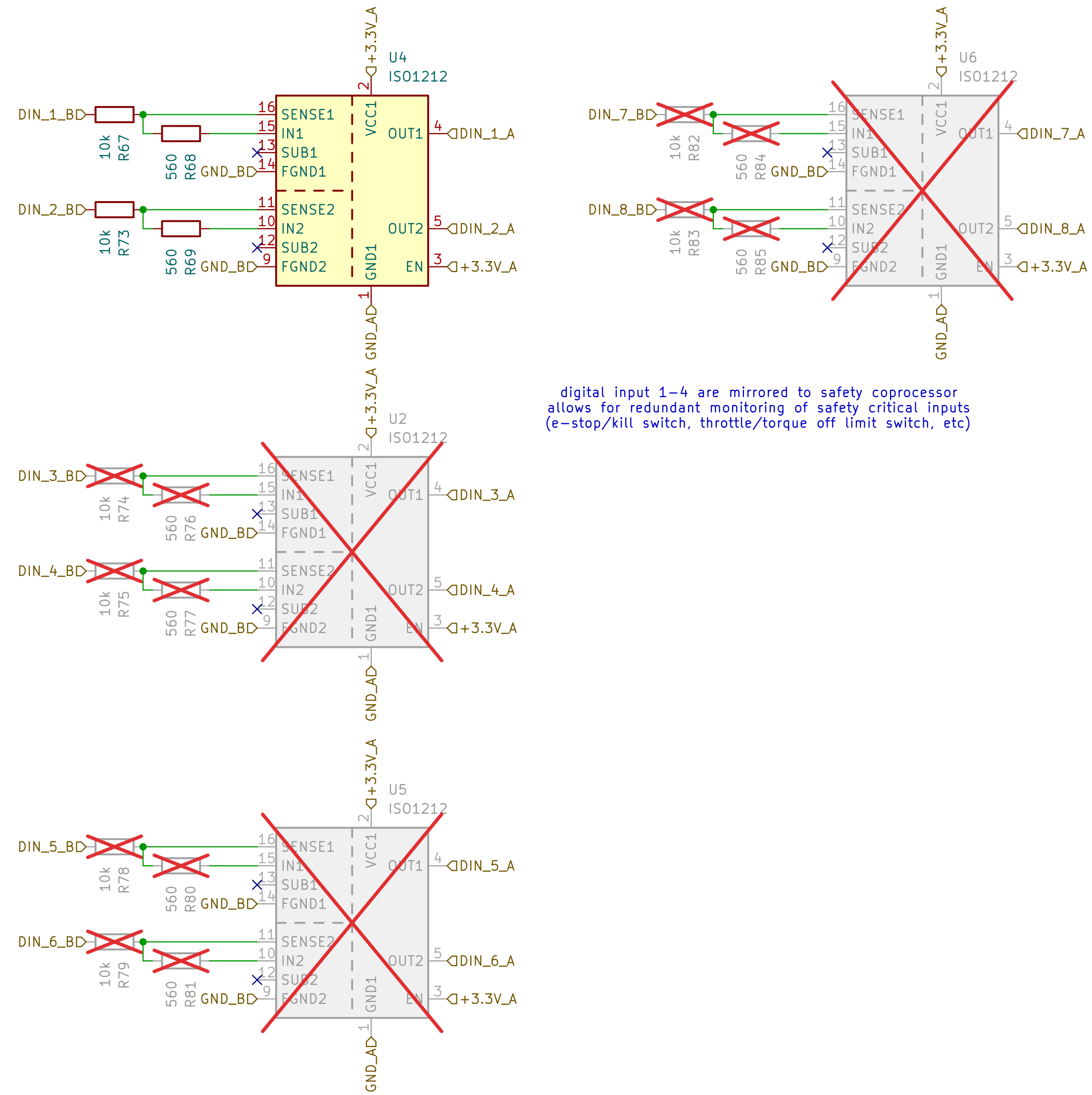
University of California, Santa Cruz, Corzine Lab

Sheet: /Current Sensors/

File: CurrentSensors.kicad_sch

Title: Open Inverter Platform

Size: USLetter	Date: 2026-07-07	**Rev: G**
KiCad E.D.A. 10.0.3		Id: 17/20



Prepared by Thomas Liao

University of California, Santa Cruz, Corzine Lab

Sheet: /Isolated Digital IO/

File: IsolatedDigitalIO.kicad_sch

Title: Open Inverter Platform

Size: USLetter	Date: 2026-07-07	**Rev: G**
KiCad E.D.A. 10.0.3		Id: 18/20